

CHAPTER 5

UNDERSTANDING COMPUTER FORENSIC

HISTORY

Until the late 1990s, what became known as digital forensics was commonly termed 'computer forensics'. The first computer forensic technicians were law enforcement officers who were also computer hobbyists. In the USA in 1984 work began in the FBI Computer Analysis and Response Team (CART). One year later, in the UK, the Metropolitan Police set up a computer crime unit under John Austen within what was then called the Fraud Squad.

A major change took place at the beginning of the 1990s. Investigators and technical support operatives within the UK law enforcement agencies, along with outside specialists, realised that digital forensics (as with other fields) required standard techniques, protocols and procedures. A series of conferences, initially convened by the Serious Fraud Office and the Inland Revenue, took place at the Police Staff College at Bramshill in 1994 and 1995, during which the modern British digital forensic methodology was established.

In the UK in 1998 the Association of Chief Police Officers (ACPO) produced the first version of its *Good Practice Guide for Digital Evidence* (Association of Chief Police Officers, 2012). The ACPO guidelines detail the main principles applicable to all digital forensics for law enforcement in the UK.

INTRODUCTION

The application of computer for investigating computer-based crime has led to development of a new field computer forensics. The goal of computer forensics is to perform a structured investigation on a digital system. Forensics science is the application of science to law and it is ultimately defined by use in court.

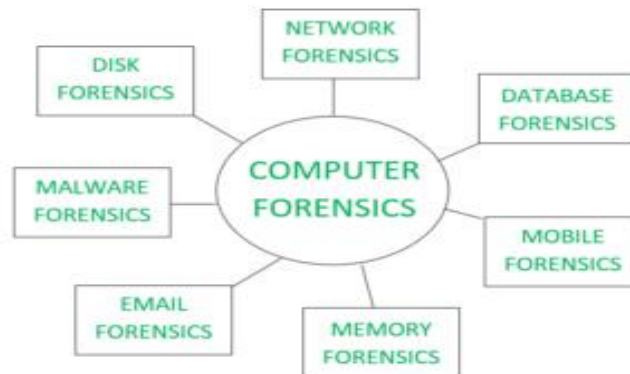
Computer forensics is primarily concerned with the systematic 'identification', 'acquisition', 'preservation' and 'analyses of digital evidence after an unauthorized access to computer or unauthorized use of computer has taken place, while the main focus of computer security is the prevention of unauthorized access to computer systems as well as maintaining confidentiality, integrity and availability of computer systems.

DIGITAL FORENSICS SCIENCE

- Is the application of analyses techniques to the reliable and unbiased collection, analysis, interpretation and presentation of digital evidence.
- It is generally considered to be related to the use of analytical and investigative techniques to identify, collect, examine and preserve evidence which is magnetically stored or encoded.
- The objective of cyber forensics is to provide digital evidence of a specific or general activity.

Computer forensics: it is the lawful and ethical seizure, acquisition, analysis, reporting and safeguarding of data and metadata derived from digital devices which may contain information. It is the collection of techniques and tools used to find evidence in a computer.

Digital forensics: it is the use of scientifically derived and proven methods towards the preservation, collection, validation, identification, analysis, interpretation, documentation and presentation of digital evidence derived from digital sources for the purpose of facilitation or furthering the reconstruction of events found to be criminal.



CHARACTERISTICS

- Identification: Identifying what evidence is present, where it is stored, and how it is stored (in which format). Electronic devices can be personal computers, Mobile phones, PDAs, etc.
- Preservation: Data is isolated, secured, and preserved. It includes prohibiting unauthorized personnel from using the digital device so that digital evidence, mistakenly or purposely, is not tampered with and making a copy of the original evidence.
- Analysis: Forensic lab personnel reconstruct fragments of data and draw conclusions based on evidence.
- Documentation: A record of all the visible data is created. It helps in recreating and reviewing the crime scene. All the findings from the investigations are documented.
- Presentation: All the documented findings are produced in a court of law for further investigations.

The role of digital forensics:

- Uncover and document evidence and leads.
- Corroborate evidence discovered in other ways.
- Assist in showing a pattern of events
- Connect attack and victim computers
- Reveal an end-to-end path of events leading to a compromise attempt, successful or not.
- Extract data that may be hidden, deleted or otherwise not directly available.

The typical scenarios involved are:

- Employee internet abuse.
- Data leak/ data breach
- Industrial espionage
- Damage assessment
- Criminal fraud
- Criminal cases
- Copyright violation

Using digital forensics techniques:

- Corroborate and clarify evidence otherwise discovered
- Generate investigative leads for follow-up and verification
- Provide help to verify an intrusion hypothesis
- Eliminate incorrect assumptions.

NEED FOR COMPUTER FORENSICS

The information and communication Technology (ICT) advances that pervasive the use of computers worldwide. The users, businesses and organizations have to live with a constant threat from hackers who use variety of techniques and tools to break into computer systems, steal information, change data and cause havoc.

- The goal of computer forensics is to perform a structured investigation and maintain a documented chain of evidence to find out exactly what happened on a computing device and who was responsible for it.
- Digital evidence is useful in criminal investigations, particularly in solving cybercrime and digital-world crimes, such as data theft, network breaches and illicit online transactions.
- It is also used to solve physical-world crimes, such as burglary, assault, hit-and-run accidents, and even murder.
- Businesses and governments also use computer forensics to find information related to a system or network compromise, and then use these discoveries to identify and prosecute cyber attackers.
- In addition, they can use digital forensic experts and processes to facilitate data recovery in the event of a system or network failure caused by a natural or other disaster.

- Digital forensics are also used in civil litigation cases (fraud, divorce), and to investigate cases of intellectual property theft.

CYBERFORENSICS AND DIGITAL EVIDENCE.

Cyber forensics can be divided into 2 domains

1.Computer Forensics.

2.Network forensics

Network Forensics - is the study of network traffic to search for truth in civil , criminal and administrative matters to protect users and resources from exploitation, invasion of privacy and other crime fostered by the continual expansion of network connectivity.

- As compared to the physical evidence, digital evidence is different in nature because it has some unique features.
- It is much easier to change/ manipulate
- Perfect digital copies can be made without harming original.
- It is usually in the form of the image- this means that it is convenient and possible to create a defensible 'clone' of storage device

Computer forensics experts know the techniques to retrieve the data from files listed in the standard directory search, hidden files, deleted files, deleted emails and passwords, login id, encrypted files etc.

Typically, the evidences reside on computer systems, user created files, user protected files, computer created files and on computer networks. Computer systems have the following:

1.Logical file system that consists of

- File system
- Random access memory
- Physical storage media
 - Slack space
 - Unallocated space

2.User created files

It consists of address books, audio/video files, calendars, database files, spread sheets, emails, internet bookmarks, documents etc.

3.computer created files

It consists of backups, cookies, configuration files, history files, log files, swap files etc.

4.Computer Networks

It consists of the Application Layer, the transportation layer, the network layer, datalink layer.

THE RULES OF EVIDENCE

According to the 'Indian Evidence Act 1872', evidence means and include:

1. All statements which the court permits or requires to be made before it by witnesses, in relation to matters of fact under inquiry are called oral evidence.
2. All documents that are produced for the inspection of the court are called documentary evidence.

There are number of contexts involved in actually identifying a piece of digital evidence:

Physical context: It must be definable in its physical form i.e. it should reside on a specific piece of media

Logical context: It must be identifiable as to its logical position i.e. where does it reside relative to the file system.

Legal context: We must place the evidence in the correct context to read its meaning.

Forensics Analysis of E-Mail

- It helps establish the authenticity of an E-Mail when suspected.
- E-Mails -- the most common means of communication.
- The subject of forensics analysis for "digital evidence."

E-Mail System

The hardware and software that controls the flow of E-Mail.

Components

1. E-Mail server
2. E-Mail gateway

The path taken by digital evidence can be conceptually depicted as shown in Fig.

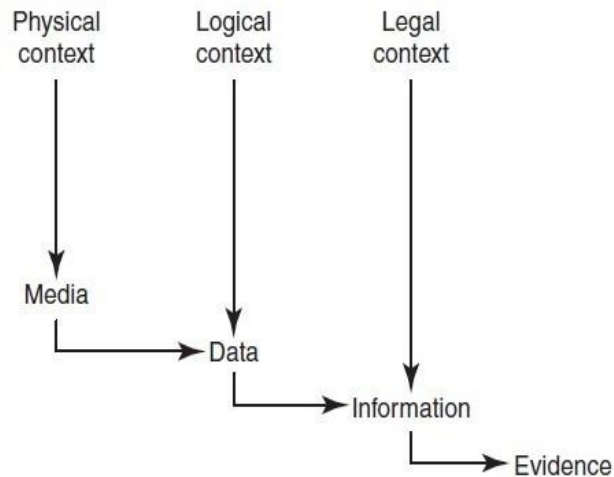


Figure 2 | Path of the digital evidence.

RFC2822

- ✓ Internet Message Format
- ✓ Several formats of valid E-Mail addresses: joshi@host.net, john@[10.0.3.19], “Joshi Ganesh”@host.net or “Joshi Ganesh”@[10.0.3.19]
- ✓ Many E-Mail address validators on the Web fail to recognize some of those valid E-Mail addresses
- ✓ RFC2822 standard applies only to the Internet Message Format
- ✓ Some of the semantics of message contents contains no specification of the information in the envelope

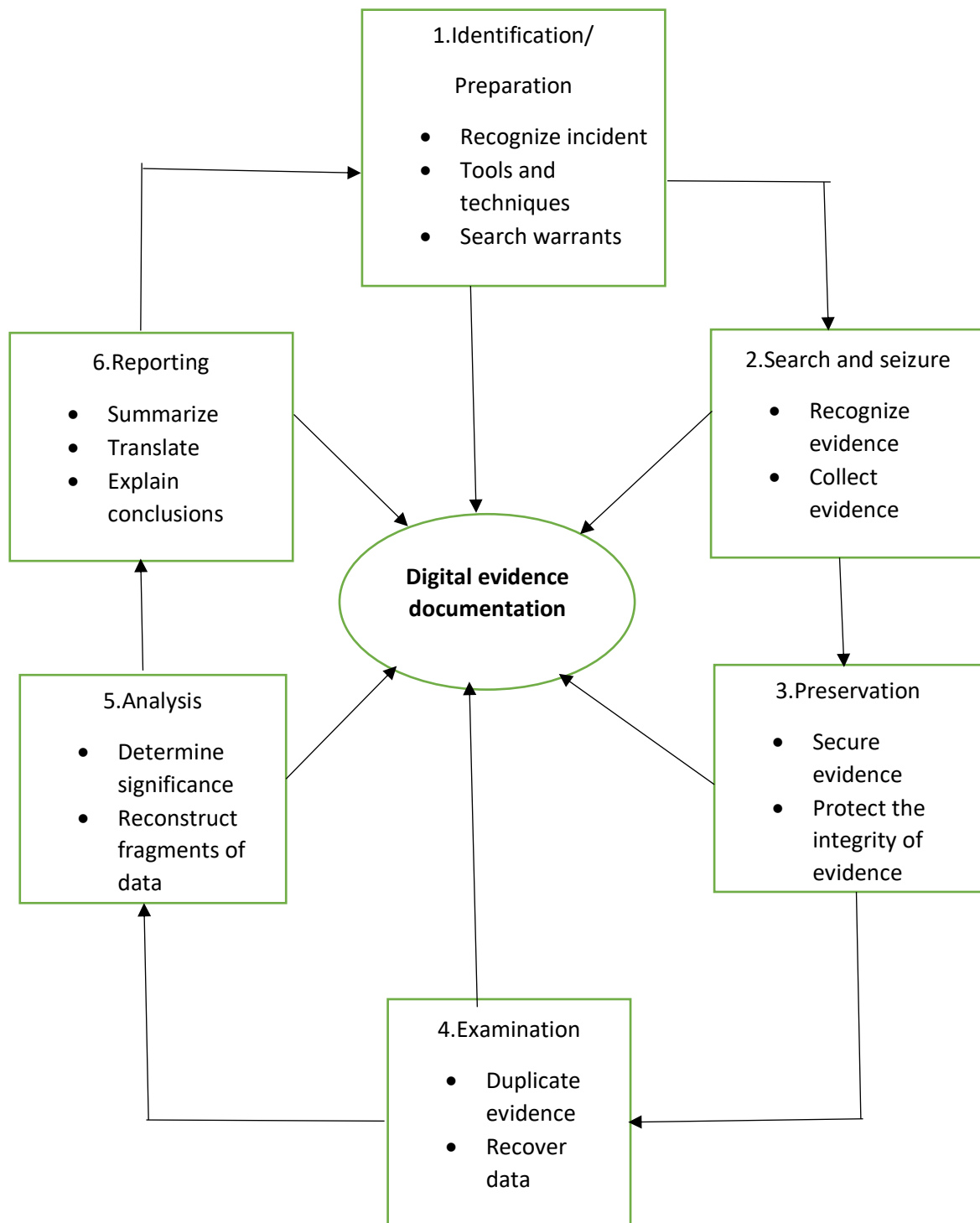
DIGITAL FORENSICS LIFE CYCLE

The law enforcement must know how to recognize, seize, transport and store original digital evidence to preserve it for forensic examination.

The cardinal rules to remember are that evidence:

- Is admissible
- Is authentic
- Is complete
- Is reliable
- Is understandable and believable

THE DIGITAL FORENSIC PROCESS



The forensic lifecycle involves the following phases:

- 1.Preparation and identification
- 2.Collection and recording
- 3.Storing and transportation
- 4.Examination/ Investigation
- 5.Analysis, interpretation and attribution
- 6.Reporting
- 7.Testifying

The process involves the following activities

1.Prepare

Case briefings, engagement terms, interrogatories, spoliation prevention, disclosure and discovery planning and requests.

2.Record

Drive imaging, indexing, profiling, search plans, cost estimates, risk analysis.

3.investigate

Triage images, data recovery, keyword searches, hidden data review, communicate, iterate.

4.Report

Oral vs written, relevant document production, search statistic reports, case log reporting.

5.Testify

Testimony preparation, presentation preparation, testimony.

PREPARING FOR THE EVIDENCE AND IDENTIFYING THE EVIDENCE

- In order to be processed and applied, evidence must first be identified as evidence.
- Consider that every sequence of events within a single computer might cause interactions with files and file systems in which they reside.
- If the evidence cannot be identified as relevant evidence, it may never be collected or processed at all.

Collecting and Recording Digital Evidence

- Digital evidence can be collected from many sources include comouters, cell phones, digital cameras, USB etc.
- Most digital information is easily changed, once changed it is usually impossible to detect that a change has taken place.
- A common practice to calculate a cryptographic hash of an evidence file and to record the hash elsewhere.
- Collecting volatile data requires special technical skills.
- If the machine is suspected of being used for illegal communications, such as terrorist traffic, not all the information must be stored on the hard disk.

Storing and Transporting Digital Evidence

- Image computer media using a write-blocking tool to ensure that no data is added to the suspect device.
- Establish and maintain the chain of custody
- Document everything that has been done
- Only use tools and methods that have been tested and evaluated to validate their accuracy and reliability.
- Original copies are kept in a secure location to act as the original evidences that is introduced into legal proceedings.

Examining/ Investigating Digital Evidence

- In an investigation in which the owner of the digital evidence has not given consent to have his or her media examined.
- Special care must be taken to ensure that the forensics specialist has the legal authority to seize, copy and examine the data.
- Traditionally, computer forensics investigations were performed on data at rest, example the content of hard disk. This can be thought of as a “**dead analysis**”.
- The process of creating an exact duplicate of the original evidentiary media is often called “**Imaging**”.
- Using a standalone hard drive duplicator or software imaging tools such as DCFLdd, IXimager or Guymager, the entire hard drive is completely duplicated.

Analysis, Interpretation and Attribution

- Analysis. Interpretation and attribution of evidence are the most difficult aspects encountered by most of the forensics analysts.
- Basically, all digital evidence must be analysed to determine the type of information that is stored upon it.
- Specially tools are used that can display information in a format useful to investigators.
 - Access Data FTK
 - Guidance software Encase

- It also includes a manual review of material on the media.
- Example-OS specific investigation is reviewing the windows registry.
- The investigators' objective is to look for suspect information, discovering and cracking passwords, performing keyword searches for crime related, extracting email etc.

Types of Digital Analysis

Each of these layers has its own analysis techniques and requirements. Example of common analysis are as follows:

- **Media Analysis**
It is analysis of the data from the storage device. This analysis does not consider any partitions or other operating system specific data structures.
- **Media Management Analysis**
It is analysis of the management system used to organize media. It typically involves partitions and may include volume management or redundant array of independent disks systems that merge data from multiple storage device into single virtual storage device.
- **File system analysis**
It is analysis of the file system data inside a partition or disk. It involves processing the data to extract the contents of a file to recover the contents of a deleted file.
- **Application Analysis**
It is analysis of the data inside a file. Files are created by users and applications.
- **Network Analysis**
It is analysis of data on a communication network, network packets can be examined using the OSI model to interpret the raw data into an application-level system.
- **Image analysis**
Digital images are the target of many digital investigation because some are contraband.
- **Video analysis**
Is used in security cameras and in personal video cameras and webcams. It examines the video for the identification of objects in the video and the location where it was shot.

Reporting

Once the analysis is complete, a report is generated. The report may be in a written form or an oral testimony or it may be a combination of the two. After extracting and analysing the evidence collected, the results may need to be presented before a wide variety of audience including law enforcement officials, technical experts, legal experts, corporate management etc.

The following are the broad-level elements of the report:

- Identity of the reporting agency
- Case identifier or submission number
- Case investigator
- Identity of the submitter
- Date of receipt
- Date of report
- Identity and signature of the examiner
- Results/conclusions

Testifying

This phase involves presentation and cross examination of expert witnesses. Depending on the country and legal frameworks in which a cybercrime case is registered, certain standards may apply with regard to the issues of expert witnesses.

- Testimony is based on sufficient facts or data.
- The testimony is the product of reliable principles and methods.
- The witness has applied the principles and methods reliably to the facts of the case.

CHAIN OF CUSTODY CONCEPTS

A trail can be lost quickly if opposing counsel can show that the evidence chain of custody was violated. Attorneys and law enforcement filled out a chain of custody form that showed who had handled the evidence and the dates and times.

To avoid the risk of mishandling evidence, proper chain of custody procedure should always be strictly observed. It also ensures that digital fingerprints match up all stages of investigation and that documentation of all evidence artifacts, as well as maintaining case logs for all evidence related activities is performed.

Collecting information regarding the environment and use of the computer or machine under investigation, in an attempt to answer questions such as the following prior to the arrival of the forensics investigator.

- Who has access to the machine?
- What level of authorization did all of those individuals having access to the machine have?
- What was the machine used for?

- What external devices did the machine connect to or interact with?
- Which and how many servers did the machine touch?
- Where and how will you store and safeguard the machine and the evidence after seizure?

The evidence or property custody should include the following information:

1. Name or initials of the individual collecting the evidence
2. Each person or entity subsequently having custody of it
3. Dates on which the evidence items were collected or transferred
4. Department name and case number
5. A brief description of the item seized.

NETWORK FORENSICS

Network forensics is a sub-branch of digital forensics relating to the monitoring and analysis of computer network traffic for the purposes of information gathering, legal evidence, or intrusion detection. Unlike other areas of digital forensics, network investigations deal with volatile and dynamic information

Network forensics is a specialized field that analyse network traffic and data to uncover the source of security incidents and malicious activities. It's a subfield of digital forensics that can be used in both criminal and civil cases.

- Open networks can be the source of many network-based cyberattacks.
- In 1997, Marcus Ranum coined the term 'wireless forensics'.
- In recent survey- Cop-Tech forum in a leading IT firm revealed that 50% of the Wi-Fi internet connection in the city continued to be unprotected.

Network forensics can help with:

- Incident response: Understanding how breaches occurred, assessing the impact, and implementing measures to prevent future incidents
- Compliance: Ensuring data integrity and security to meet regulatory requirements
- Investigating cybercrime: Identifying compromised devices, tracing attack vectors, and determining the extent of data exfiltration

Network forensics involves:

- Examining traffic data, logs, and other data
- Identifying and preserving digital evidence
- Following appropriate steps to ensure that all evidence is collected and preserved

This process involves capturing all data moving over Wi-Fi network and analysing network events to uncover network anomalies, discover the source of security attacks and investigate breaches on computers and wireless networks.

When performing wireless forensics, the security analyst must follow the same general principles that apply to computer forensics: identify, preserve and analyse the evidence to impartially report the findings and conclusions.

IMPORTANT QUESTIONS

- 1.Explain computer forensics and its characteristics.
- 2.Explain the need of computer forensic
- 3.Explain in detail the lifecycle of digital forensics.
- 4.Explain types of digital analysis of computer forensics.
- 5.Explain the activities involved in process of computer forensics.
- 6.Explain chain of custody concepts.
- 7.Explain about network forensics.
- 8.Explain about digital forensics science.